

The background is a dark, blue-toned illustration of a digital cityscape. It features a large, stylized eye in the center, surrounded by various graffiti tags such as 'CONVI', 'HACK ALL THE THINGS', and 'BIG BROTHER IS WATCHING'. There are also patterns of blue dots and lines, suggesting a network or data flow.

Digital Leviathan

Nation-State Big Brothers (from huge to little ones)

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Las Vegas, August 11th 2018

AGENDA

INTRO

SOURCES

OUTCOMES

CONCLUSION

INTRO

“We should give our obedience to an unaccountable sovereign otherwise what awaits us is a ‘state of nature’ that closely resembles civil war – a situation of universal insecurity”.

Thomas Hobbes



The claim

Terrorism, cybercrime, foreign espionage, among other are examples that support politicians claims for increase state power over online life.



Opposing evidence

The current use of cyber offensive tools are aimed to political objectives rather than public safety

INTRO

1

Espionage

Stealing of information and/or personal data (for achieving political purposes, i.e. opposition party, NGO, ethnic minorities)

2

Surveillance / Eavesdropping

Untargeted violations of privacy (monitoring of behavior, activities, or other changing information of people online)

3

Censorship

Block specific applications or technologies; filtering and blocking of websites; manipulation of content or traffic manipulation; violations of user rights.

SOURCES

SOURCES

1

APT Reports

- *Total of 758 reports/blog posts from vendors, NGO, CSIRT and universities,*
 - *The dataset has bias from a western perspective (close to 80%)*
-
- *Focus on attacks targeting NGO, political groups, media outlets, or opposition was considered as an indicator of state misbehavior*

2

Leaks from spyware providers

- *Two big providers of surveillance solutions Hacking Team and Gamma Group,*
 - *Data is available from multiple sources (Wikileaks has a good search platform)*
-
- *It was possible to identify who acquired it (LEA, military, intelligence, etc)*

3

Acquisition of technology

- *Surveillance and/or intrusion technologies reported from multiple sources*
 - *Purchases made by different countries (potential cyber capabilities)*
-
- *Buggedplanet.info and Surveillance Industry Index (SII) were helpful sources*

SOURCES

4

Censorship

- *Freedom House, OONI, Google, Reporters without Borders and OpenNet provide evidence of some level online censorship,*
- *Blocking applications, technologies, traffic; filtering and blocking of websites; violations of user rights*

- *Based on a western view of freedom of speech (an individual or a community to articulate their opinions and ideas without fear of retaliation, censorship or sanction)*

5

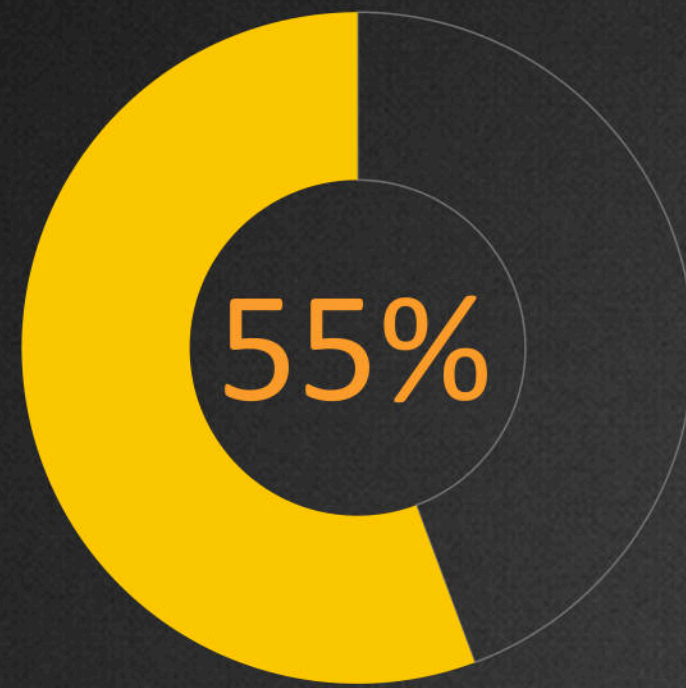
Transparency Reports

- *Transparency Reports issued by major social networks and content providers*
- *Facebook, Twitter, Google, Yahoo, Apple, LinkedIn, Snapchat, Tumblr, Dropbox, Wiki, Microsoft, and WordPress*

- *Provides extra detail on the intent of Nation-States use of social media for surveillance*

OUTCOMES

OUTCOMES – STATE CYBER ATTACK



Documents had some level of attribution



Single APT Groups and/or Campaigns



Considered state-sponsored attacks

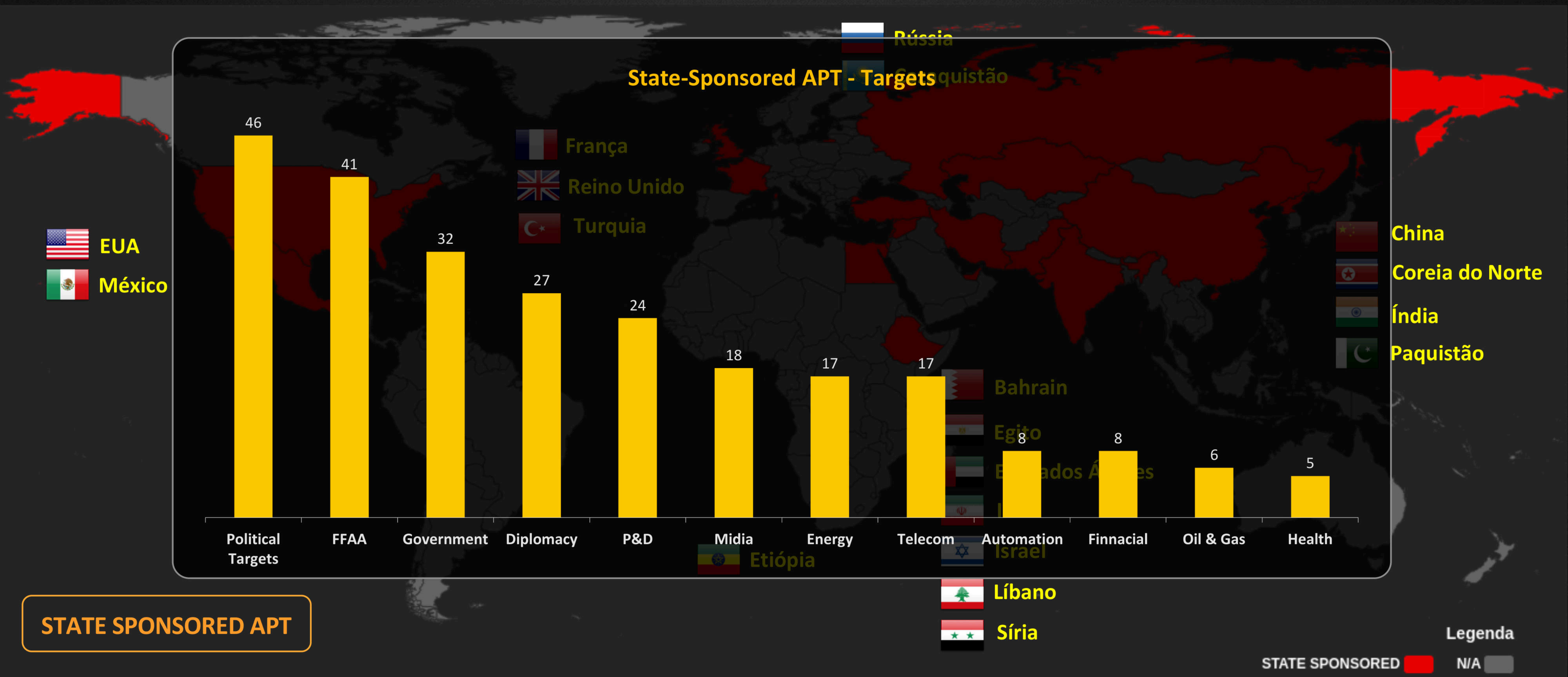


Countries attributed with a state-sponsored APT



Extensive use of Python3 (NLTK) and regular expressions for processing documents / posts

OUTCOMES – STATE CYBER ATTACK



STATE SPONSORED APT

OUTCOMES – CYBER OFFENSIVE CAPABILITIES

71

Countries acquired
offensive solutions from
private vendors

57%

In 41 cases it was
possible to identify the
user/buyer

19

User/buyer was an
intelligence agency and/
or armed forces

29

Countries that acquired
more than one
offensive solution



Wikileaks' search platform, Buggedplanet.info, Surveillance Industry Index,
and reports from Citizen Lab / Privacy International / Freedom House

OUTCOMES – CYBER OFFENSIVE CAPABILITIES

Major Private Vendors



Bull / Amesys
2 countries



Cyberbit
10 countries



Dreamlab
2 countries



Gamma Group
55 countries



Hacking Team
37 countries



NSO Group
3 countries



Procera
2 countries



SS8
4 countries



Trovicor
9 countries

ACQUIRED OFFENSIVE SOLUTIONS

Legenda
SOLUÇÃO ADQUIRIDA  N/A 

OUTCOMES – CYBER OFFENSIVE CAPABILITIES

Users / Buyers



Saudi Arabia

GIP / GID / MD



Spain

CNI



Mongolia

SSSD



Singapore

IDA SGP



Azerbaijan

Azerbaijan NS



Hungary

SSNS



Malaysia

MACC / MALMI / PMO



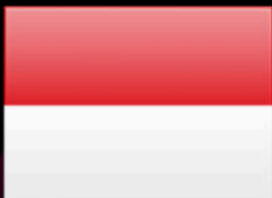
Thailand

Royal Thai Army



Bangladesh

(DGFI)



Indonesia

Lembaga Sandi Negara



Oman

Intelligence Agency



Uganda

CMI



Cyprus

Intelligence Agency



Kenya

NIS



Panama

Presidency Cabinet



Uzbekistan

NSS



Ecuador

SENAIN



Marroco

CSDN / DST



Serbia

BIA

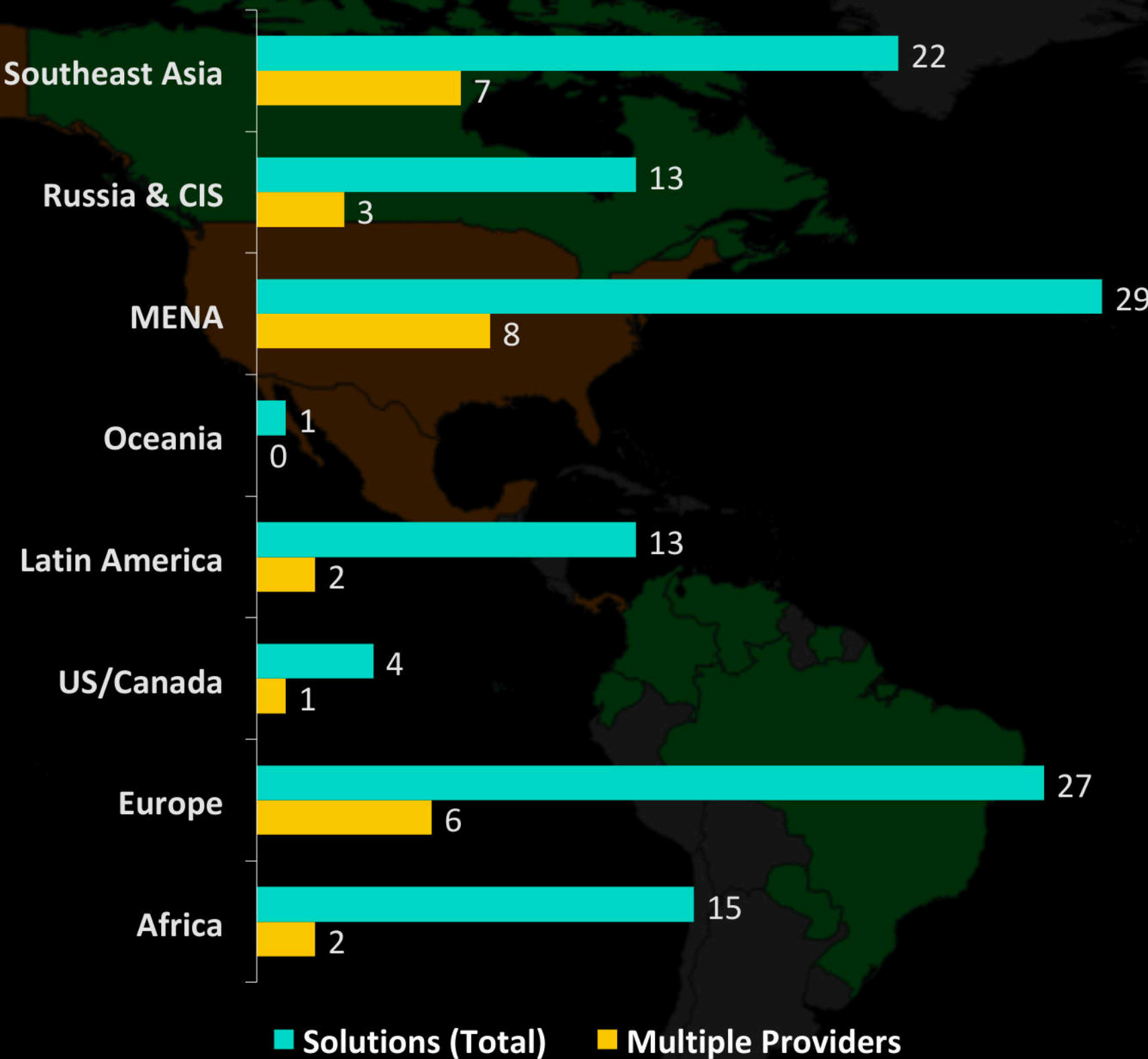
NEW THREATS

Legenda

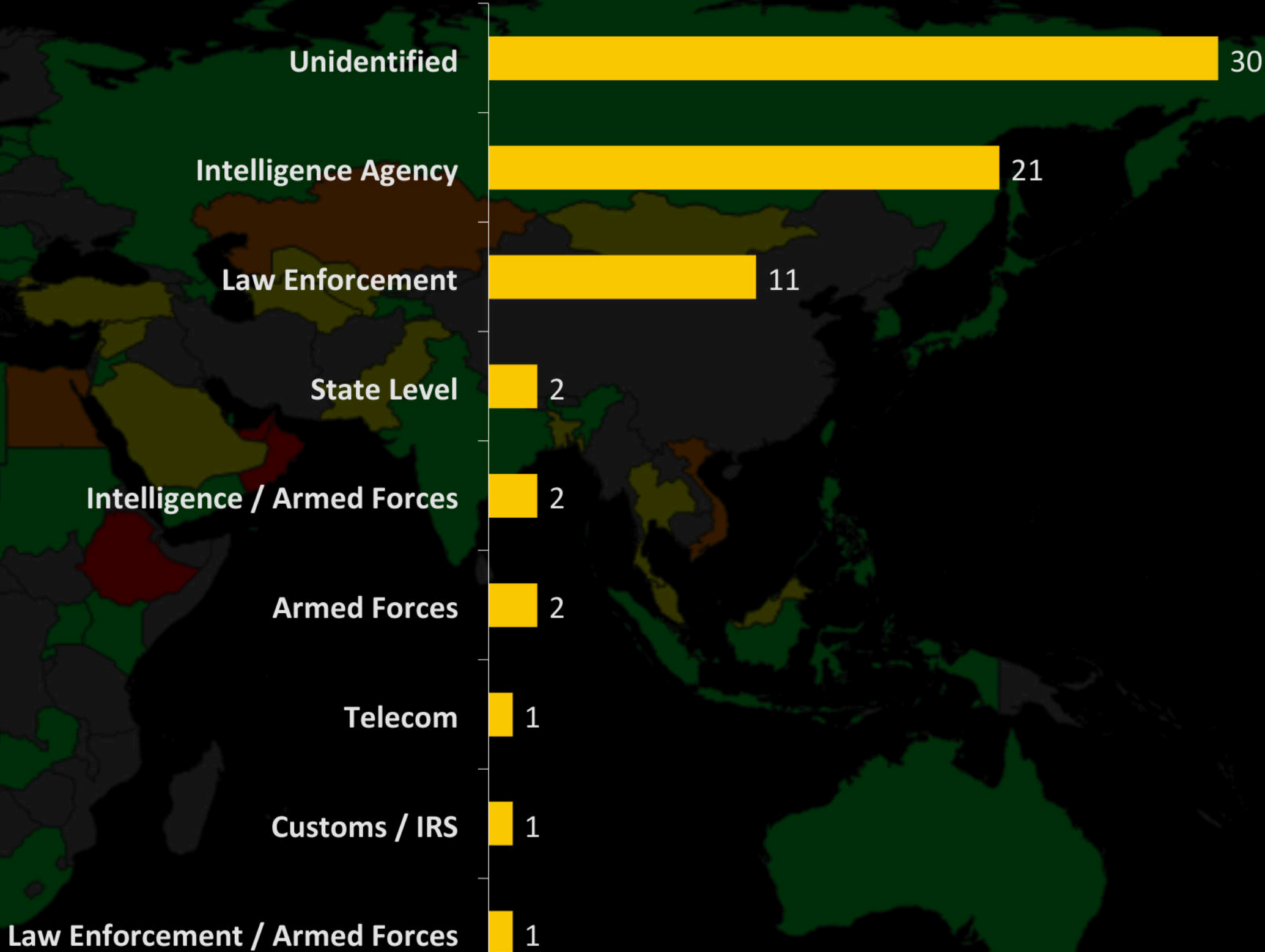
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OUTCOMES – CYBER OFFENSIVE CAPABILITIES

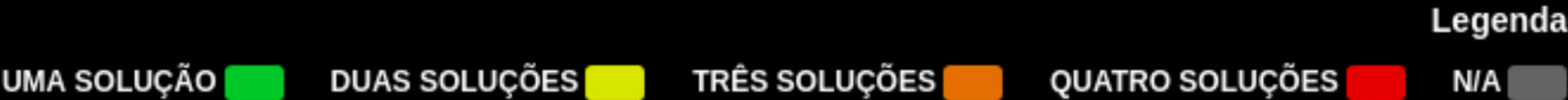
Contries with multiple solutions



User / Buyer



MULTIPLE OFFENSIVE SOLUTIONS



OUTCOMES - CENSORSHIP

40

Countries with evidence
of online censorship

42

Countries with evidence
of some level of
internet shutdown

74%

In 32 countries the
shutdown reached
national level

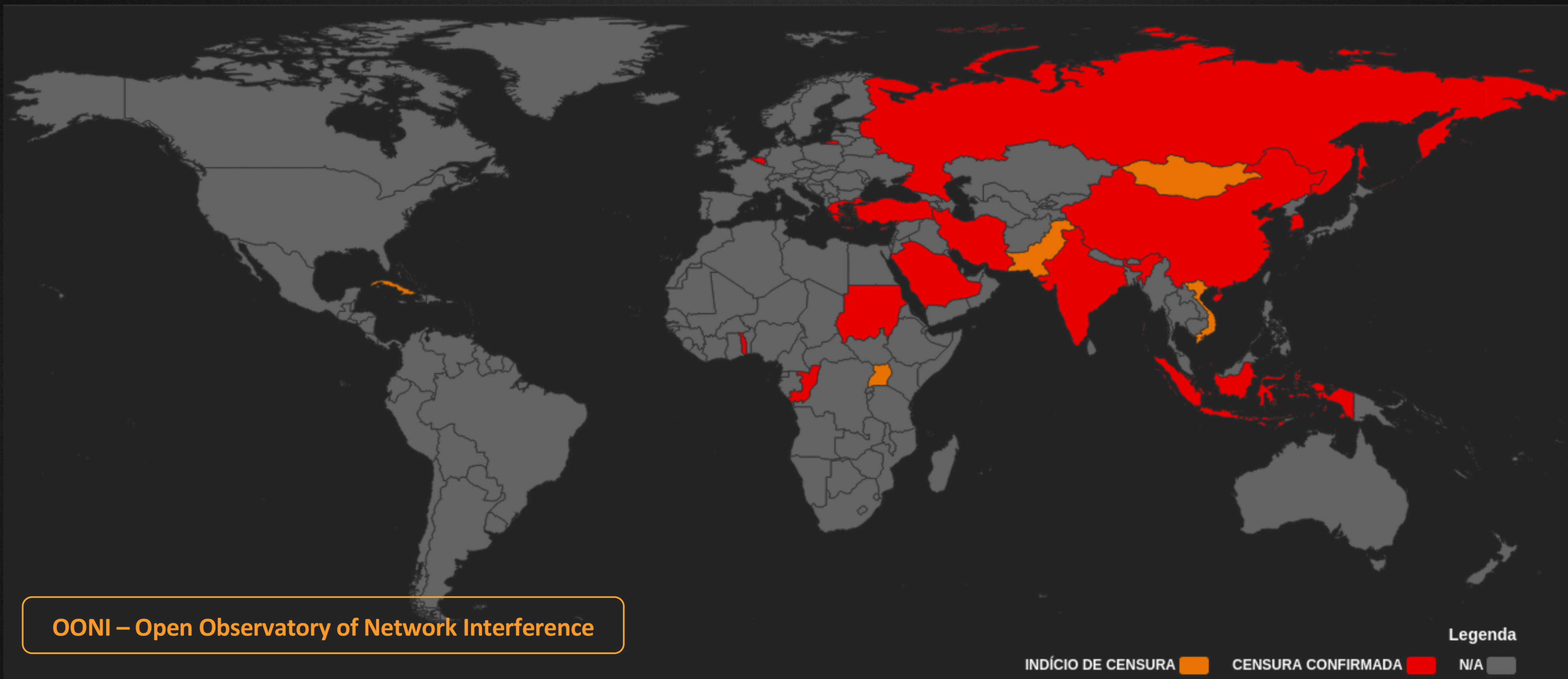
57

Countries with evidence
from two or more
sources

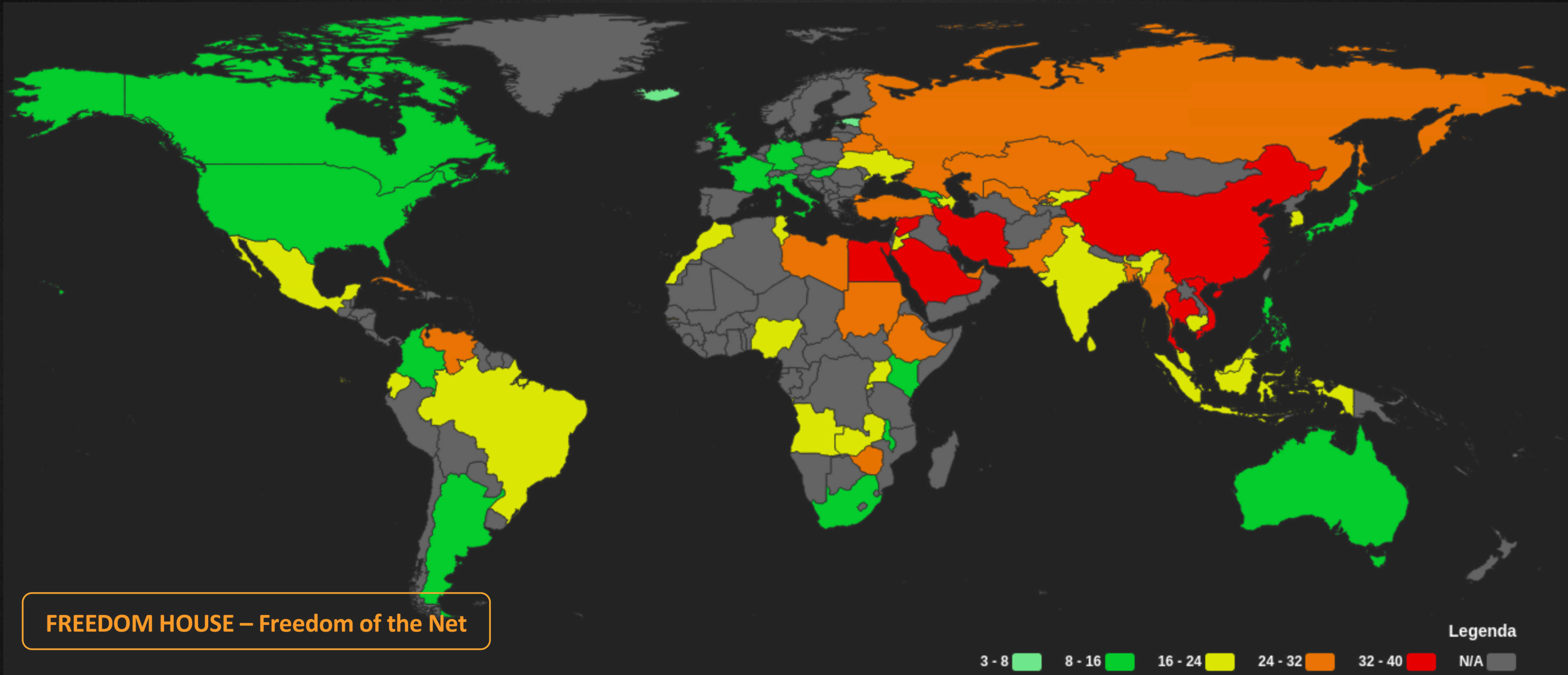
+

Reports from *Freedom House*, *OONI*, *Google*, *Reporters without Borders* and *OpenNet Initiative*

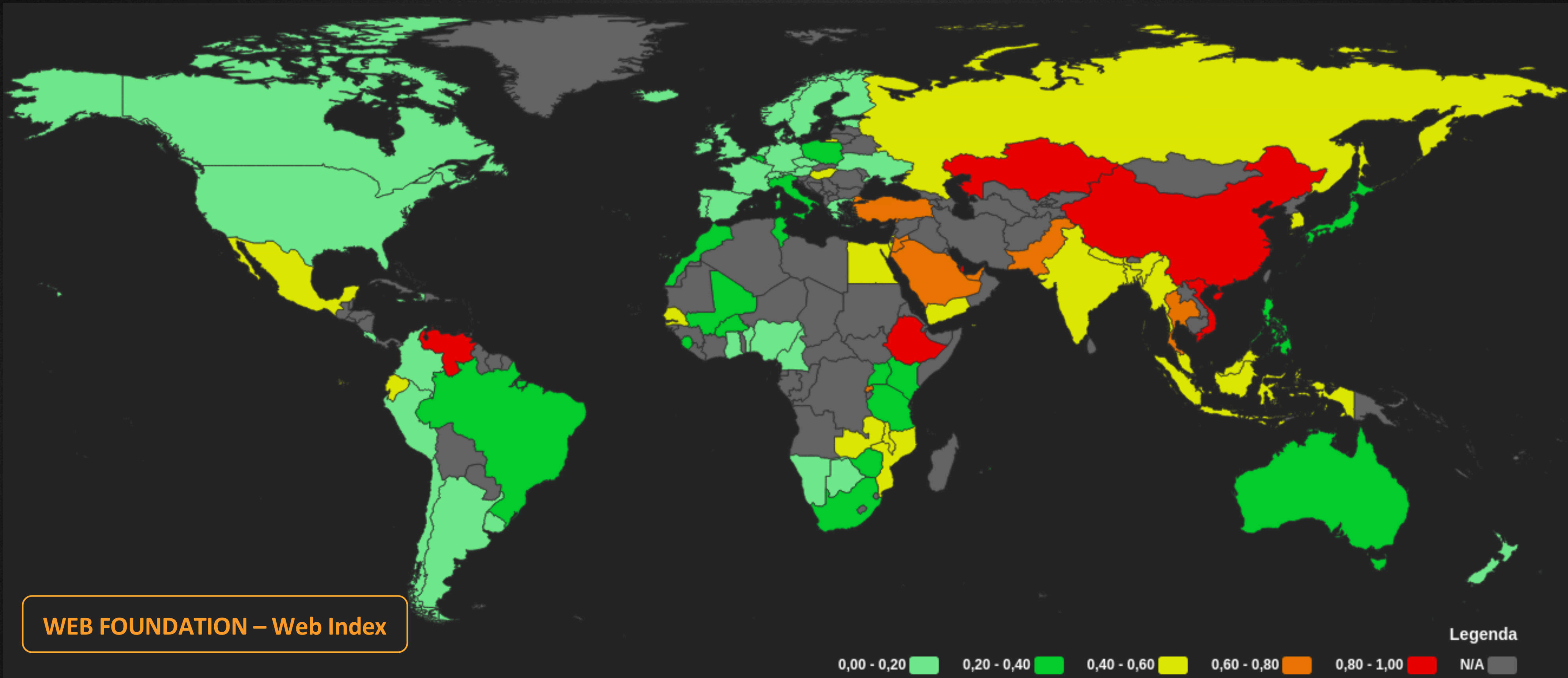
OUTCOMES - CENSORSHIP



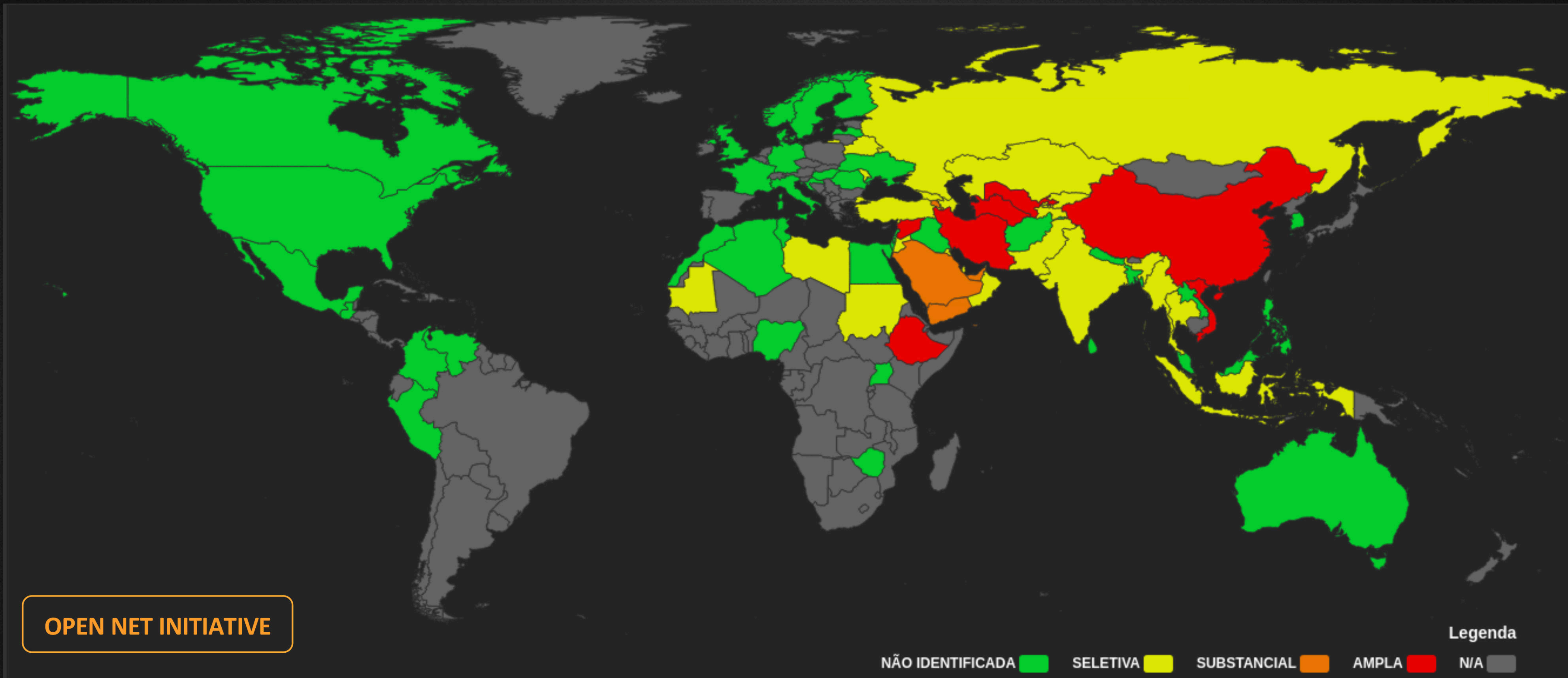
OUTCOMES - CENSORSHIP



OUTCOMES - CENSORSHIP



OUTCOMES - CENSORSHIP



OPEN NET INITIATIVE

#OpStopBlocking – Brazil 2016



Censorship and Shutdowns – Multiple Sources

	Saudi Arabia All sources		China All sources
	Vietnam All sources		Bahrain 4 sources
	Turkey 4 sources		Pakistan 4 sources
	Iran 4 sources		Ethiopia 4 sources
	United Arab Emirates 3 sources		

OUTCOMES – TRANSPARENCY REPORTS

70

Companies worldwide
publish transparency
reports

10

Major content providers
covered in this analysis

64%

World average requests
where some data was
produced (2017)

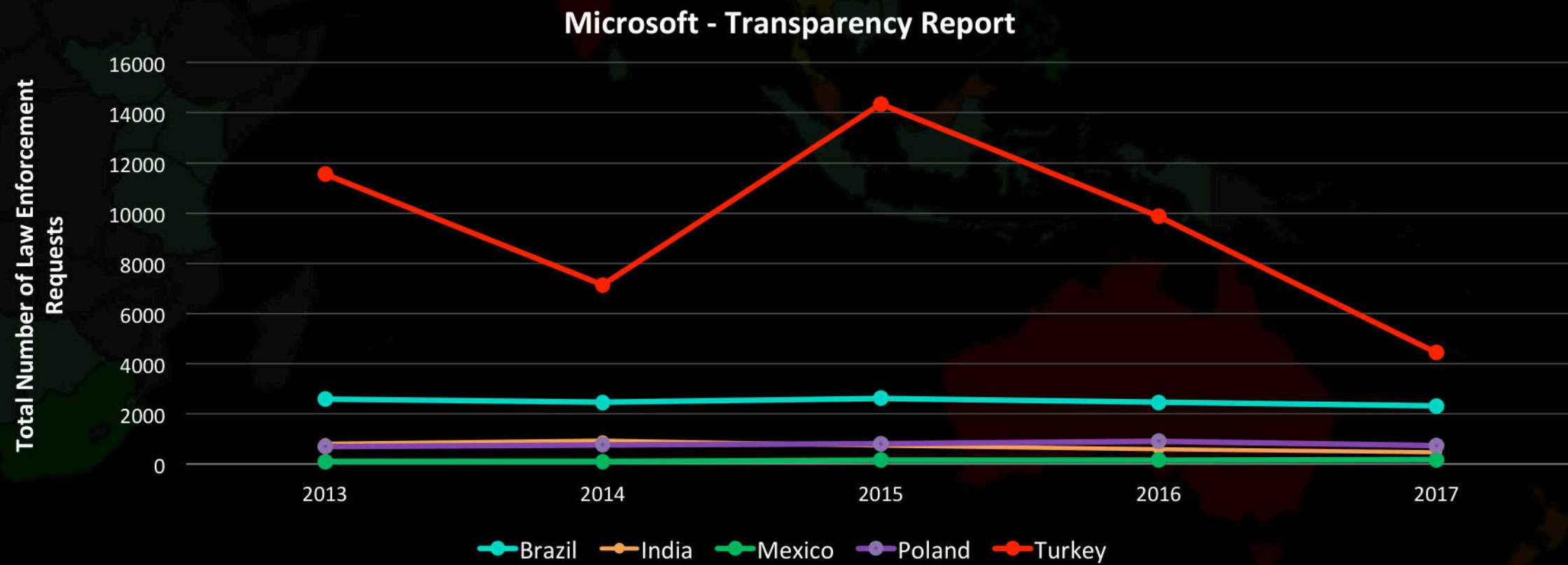
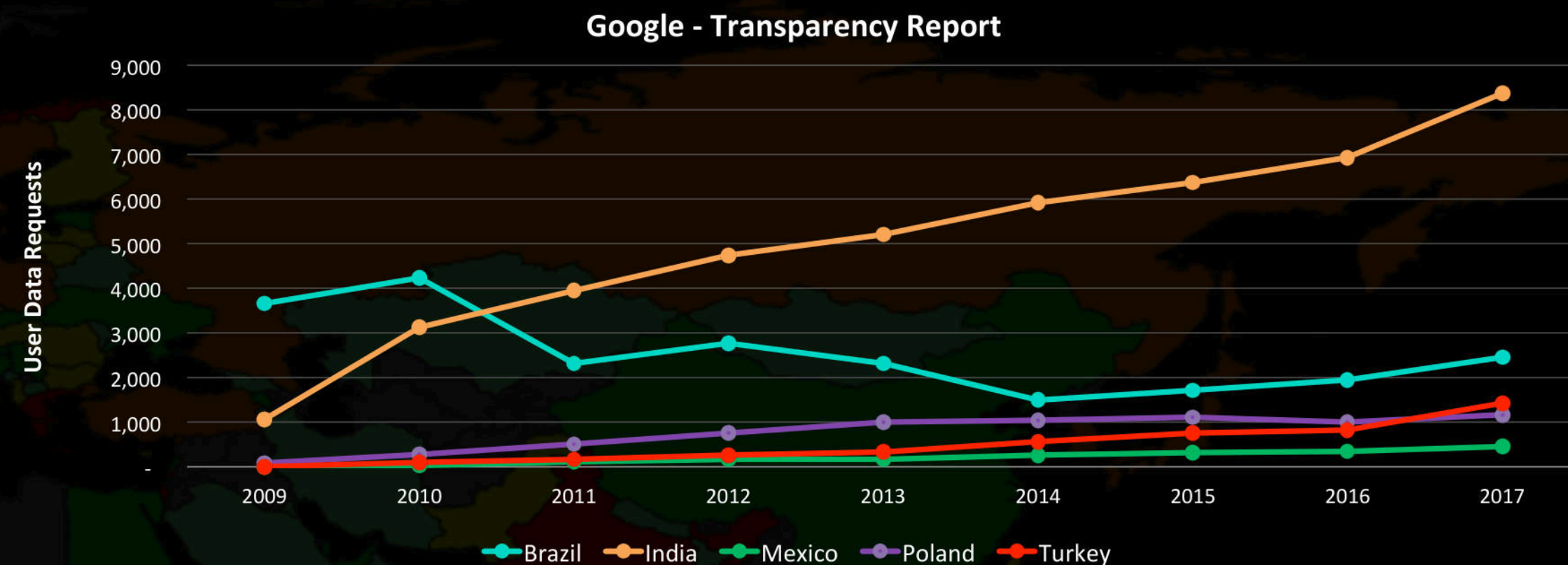
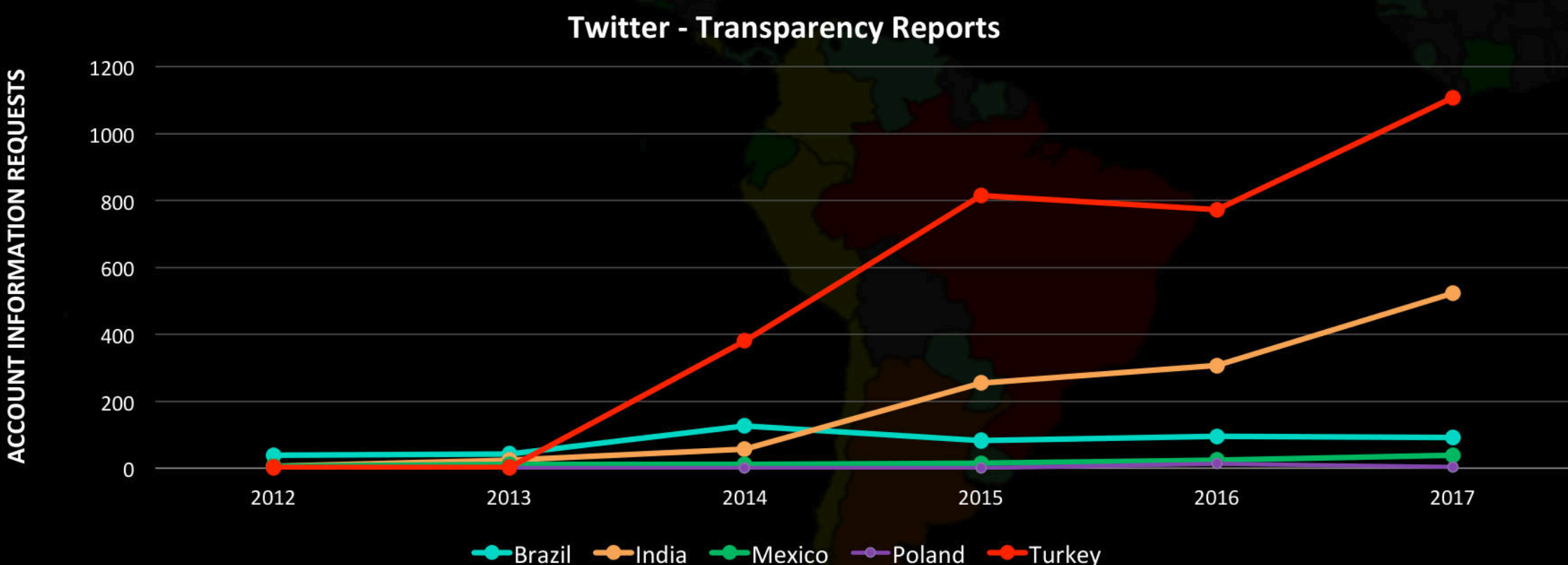
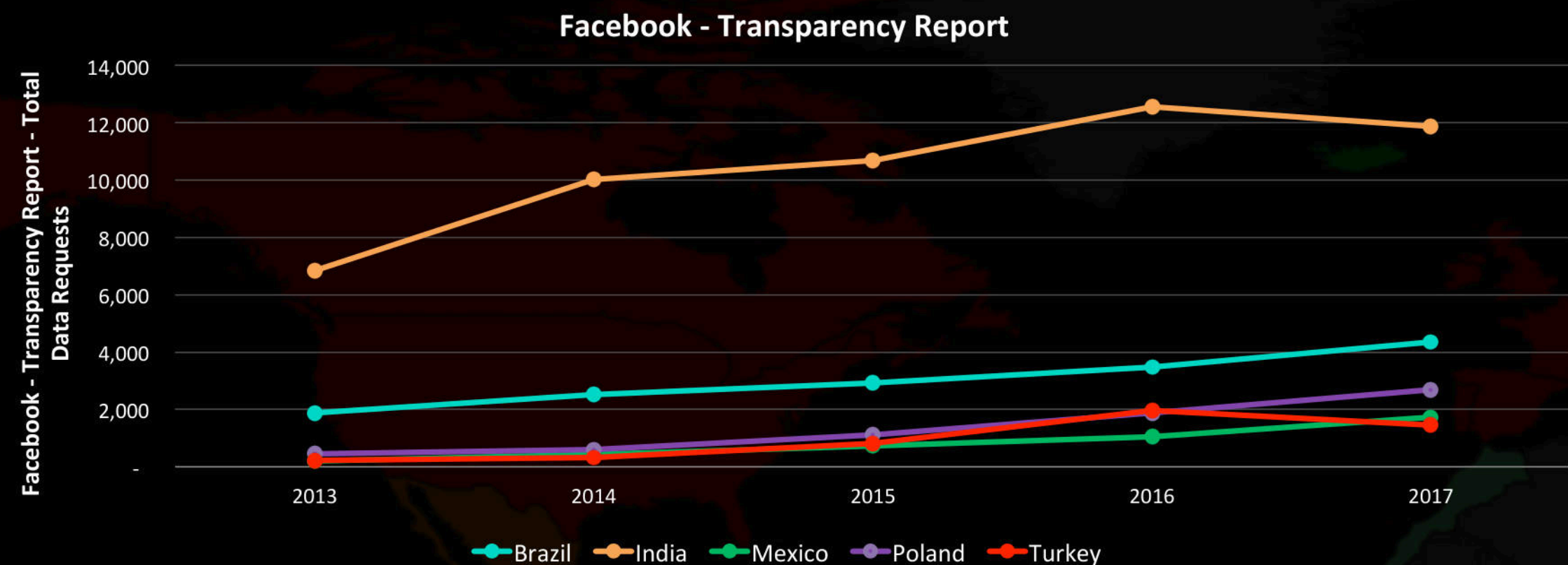
125

Countries have
requested information
or to remove content

+

Most transparency reports data are available in csv/json formats, unfortunately some only PDF

OUTCOMES – TRANSPARENCY REPORTS

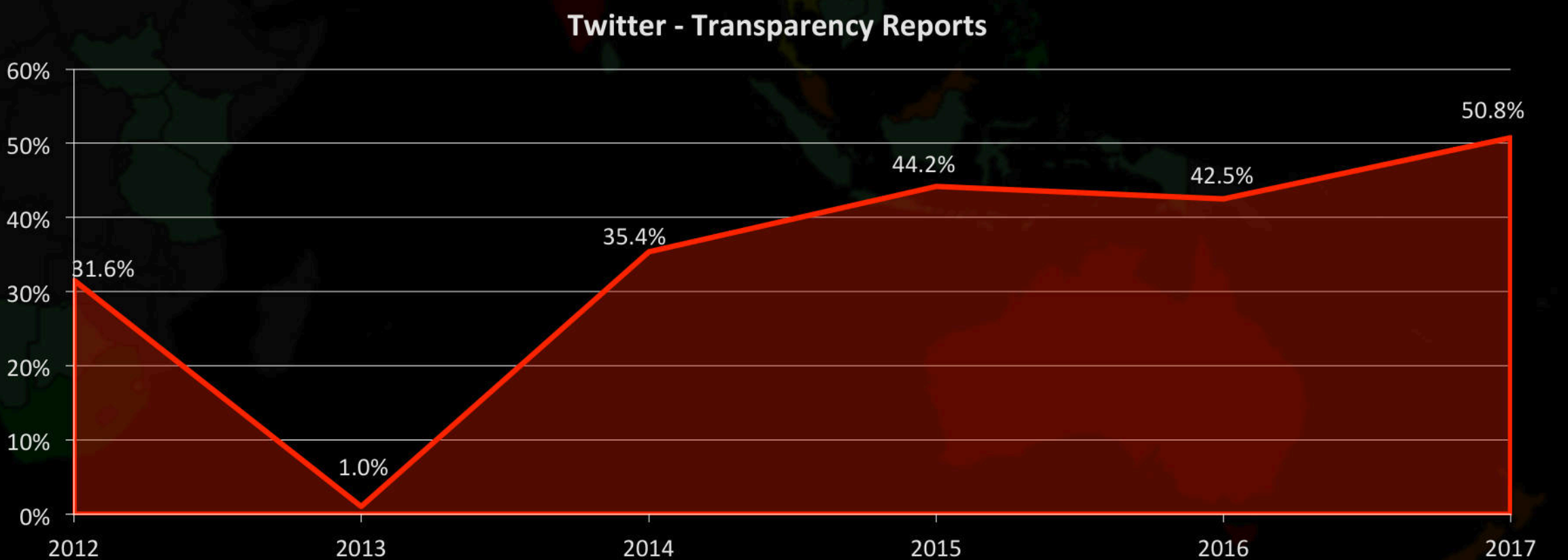
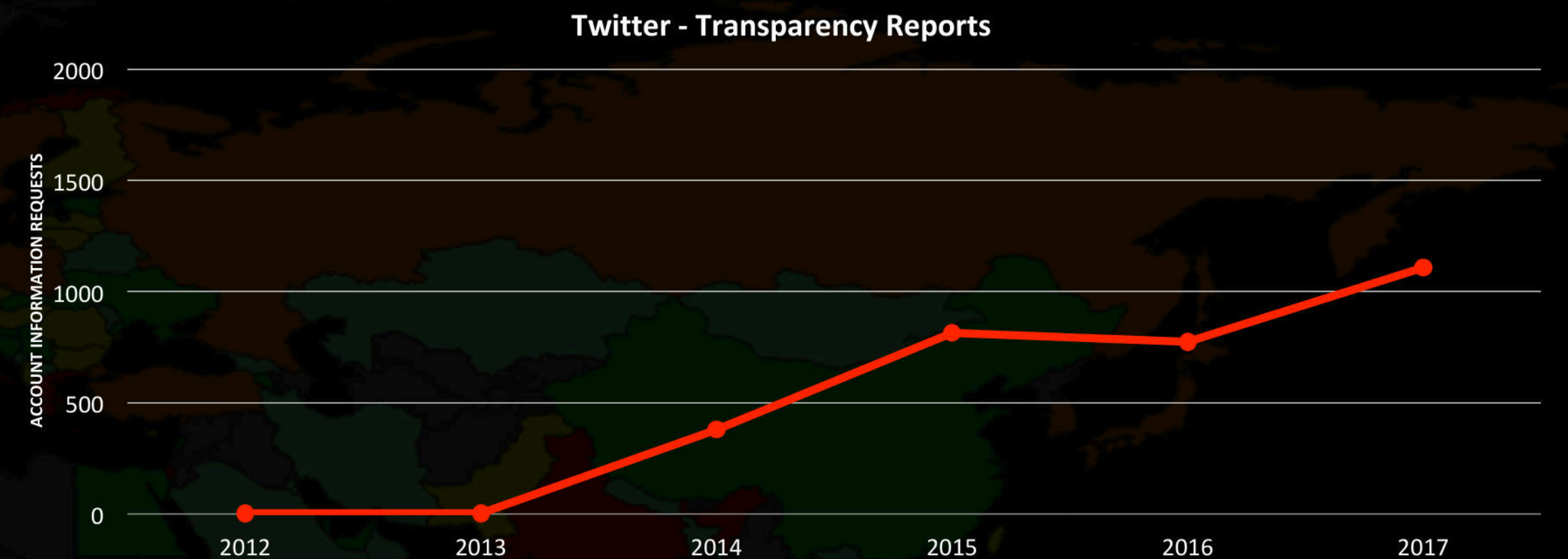
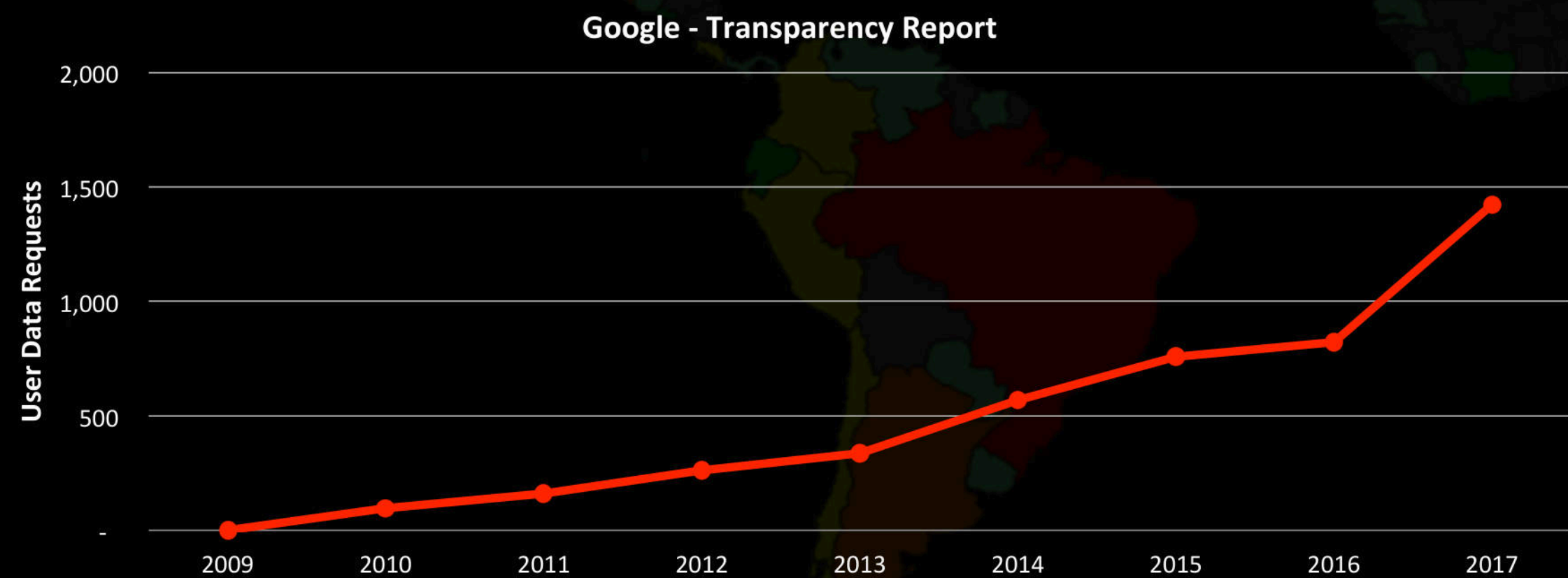
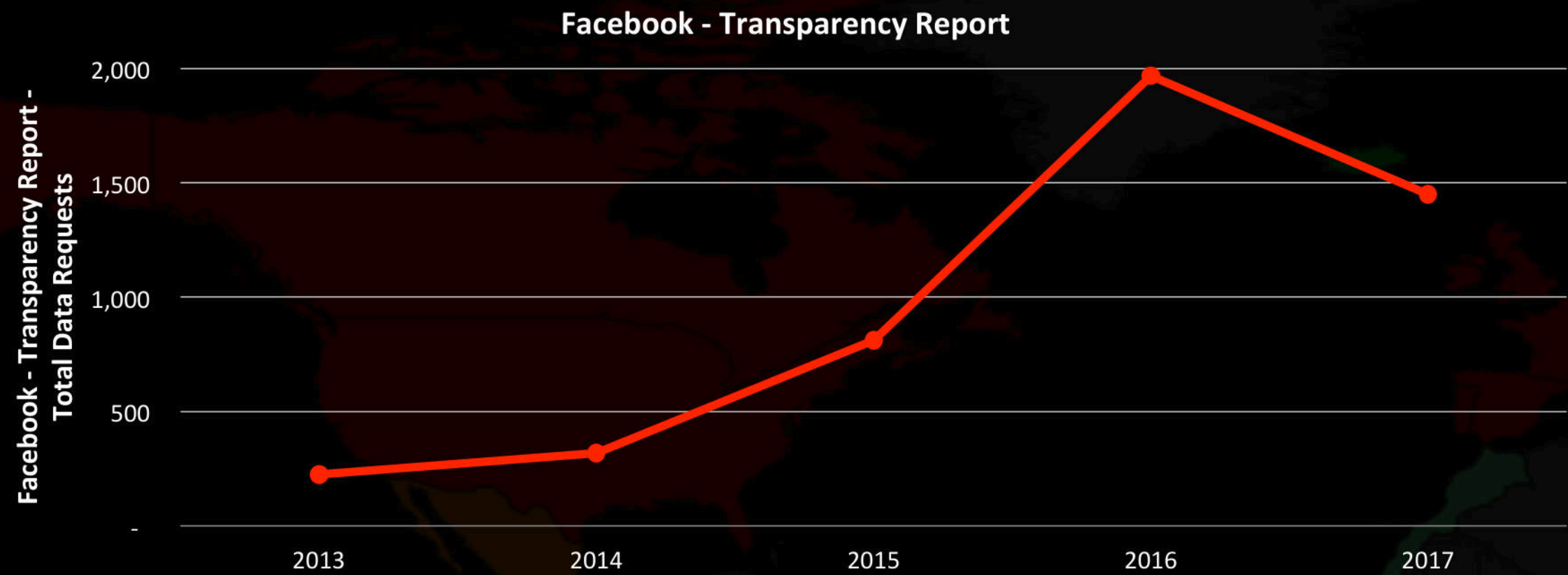


Transparency Reports

Legenda

1 - 2 2 - 4 4 - 6 6 - 8 8 - 10 N/A

OUTCOMES – TRANSPARENCY REPORTS



Transparency Reports - Turkey

Legenda



CONCLUSION

CONCLUSION

1

Cyber offensive tools being used against political targets

85 countries attacked and/or possess cyber weapons

There are evidence that 19 countries were attributed as authors/sponsors of cyber attacks.

Another 18 countries possess cyber weapons with Intelligence Agencies

At last, 34 countries acquired cyber weapons but the user/buyer is not known

All Cyber Offensive Capabilities



**92,2%
Worldwide**

OFFENSIVE CAPABILITIES

FFAA



INTELLIGENCE



STATE SPONSORED APT



LEA



CIVILIAN



UNKNOWN



N/A



Legenda

CONCLUSION

2

Online Censorship and Internet Shutdowns

The two are strongly correlated

Out of 57 countries that engaged in censorship or shutdowns, 26 did both.

Concerning social media such as Facebook, Twitter and Google, all countries have increased its yearly number of requests

Censorship/Shutdowns – Internet Users



**56,7 %
Worldwide**

CENSORSHIP & BLOCKING

Legenda

SOCIAL MEDIA



SHUTDOWNS



BOTH



CENSORSHIP



N/A



Thanks!

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